

Speech Recognition

Exercises about CFGs and NLTK

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Part 01

a) Given the following productions of a Chomsky Grammar:

$$S \rightarrow A1B$$

$$A \rightarrow 0A|\epsilon$$

$$B \rightarrow 0B|1B|\epsilon$$

Define the full Chomsky Grammar for it.

b) Give an example for a CFG producing L2.

And Regular Language L3.

Part 02

The following grammar generates the language of regular expression that look like $0^*1(0 + 1)^*$:

$G = (\{S, A, B\}, \{0,1\}, P, S)$ with P contains

$$S \rightarrow A1B$$

$$A \rightarrow 0A|\epsilon$$

$$B \rightarrow 0B|1B|\epsilon$$

Leftmost derivations always replace the leftmost non-terminal on the right-hand side of a production.

Similar, rightmost derivations replace the rightmost non-terminal.

Find the leftmost and the rightmost derivations of the following strings:

a) 00101

b) 1001

c) 00011

Part 03

Here are productions of a grammar for simple german sentences:

$$S \rightarrow NG VG$$

$$NG \rightarrow Art N \mid N$$

$$VG \rightarrow V$$

$$Art \rightarrow der \mid die \mid das \mid ein \mid des$$

$$N \rightarrow hund \mid hunde \mid haus \mid kind \mid kinder$$

$V \rightarrow \text{bellt} \mid \text{bellen} \mid \text{spielt} \mid \text{spielen}$

- a) Give 3 examples of sentences that can be derived. One should have incorrect german grammar.
- b) Give 2 reasons with examples of sentences that can not be derived.

Part 04

Use NLTK to parse the simple german sentences from exercise 03. Taking the example from

<https://www.nltk.org/book/ch08.html>

Validate a sentence like “der hund bellt” by printing the parse tree. What happens, if the sentence can not be derived?

Hint: Begin the program with

```
import nltk

german_grammar = nltk.CFG.fromstring("""
S -> NG VG
...
sent = ...

parser = nltk.ChartParser(german_grammar)
for tree in parser.parse(sent):
    print(tree)
```

Part 05

Start the Recursive Descent Parser Demo:

```
import nltk
nltk.app.rdparsers()
```

Change the grammar in rdparsers to the one for german from exercise 03. Change the textparse to ‘das kind spielt’ and show the parse tree.

Part 06

With this exercise a featured grammar is introduced.

Using `nltk.download`, make sure that the german grammar is on your computer.

It is a featured grammar. The features are annotations about grammatical issues of a word.

Parse a german sentence like 'die Katze sieht den Hund'. What is the output?

Hint for the program:

```
import nltk
from nltk import parse

cp = parse.load_parser('grammars/book_grammars/german.fcfg', trace=0)
sent = 'xyz'
tokens = sent.split()
trees = cp.parse(tokens)
for tree in trees: print(tree)
```

Find and analyze the file `german.fcfg`.

It can be read in an editor. You may change or extend the grammar.

Extend it so that other animals like Pandas or Hamsters are included.

The next step is to add verbs ...

Test the extended grammar with a sentence like "Der Hund sieht den Panda".

Part 07

Here is a simple grammar for English noun phrases:

```
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the'
N -> 'dog' | 'cat'
V -> 'chases' | 'sees'
```

Draw the parse tree for the valid sentence "the dog chases the cat".

Write the tree in bracket notation.

Part 08

Write down the steps for recursive descent parsing of the grammar and valid sentence of part 07.

Part 09

Write down the steps for shift-reduce parsing of the grammar and valid sentence of part 07.

Practical

Part 01

Implement exercise 03 (about the simple grammar for german sentences) with your chatbot. Find examples of correct sentences that are parsed. Also, find wrong sentences that are not parsed.